

– that leaves no room for internal heatsinks or extra spaces to allow cooling these components separately. Heating up of any component inside SoC would also heat up other components. So, if you are playing a game on the mobile which does not tax your processor but only the GPU, it is still the same chip which is getting hot. If all of them are put under load, the temperature would rise quickly and the voltage regulation unit would shut down the chip once the temperature threshold value is reached.

SoC versus CPU

For once, let's compare apples with oranges. We are going to do the unfair comparison of a SoC vs the CPU. Before we proceed, let's clarify the unfairness. SoCs are complete systems on a single chip, while CPUs are only processors with some cache and no RAM or Storage (modern Intel CPUs do have a graphics processor built-in though). In addition, the SoC can house DSP modules of various kinds too. It is, thus, a tad unfair comparing these two. But let us tell you – it's interesting!

Note: While it is common to address processors inside SoCs as CPUs as well, we are not doing that here. When we say CPU, we mean the processor you find in a Laptop, Desktop or a Server.

When SoC beats down the CPU

There are certain areas where CPUs can't stand up to SoCs:

1. The SoC contains the entire system and it can be tasked with anything that is achievable by a PC. The CPU, obviously, cannot get that done alone.
2. SoCs can contain DSP modules which makes them super useful. Audio synthesis, imaging, video and analog communication need DSPs which can be built right into the SoC. In many cases, the CPU might not have those capabilities, especially communications and audio synthesis systems.
3. Size is obviously an issue addressed by SoCs. Many smartwatches are full blown computers. It might be interesting to know that the Apple Watch has more processing power and memory than the system that runs Curiosity Rover on Mars. All that power, on your wrist.
4. Power consumption is something SoCs were designed to address. Mobiles, Tablets, Watches and mini computers – they all are supposed to work on as less power as possible. A part of that is because of the problems heat would pose, but it's a win for SoCs nonetheless. An SoC for a tablet would be fine with 12-15 watts doing all kinds of